

Multipurpose VTOL Drone

Hybrid Long-Range UAV for Multi-Mission Operations

OVERVIEW

The HVTOL-01 is a high-performance hybrid VTOL UAV that merges fixed-wing endurance with rotor-based flexibility. Engineered for demanding long-range operations across medical delivery, surveillance, and precision mapping — deployable from any terrain, with zero runway dependency.

TECHNICAL SPECIFICATIONS

Wingspan	1.8 – 2.5 m
Payload Capacity	2 – 5 kg
Flight Time	90 – 150 minutes
Operational Range	50 – 120 km
Cruise Speed	60 – 100 km/h
Takeoff & Landing	Vertical (VTOL) — No Runway Required
Navigation	Autonomous Waypoint + Manual Override
Airframe	Carbon Fiber Composite
Propulsion	Hybrid BLDC Motors + ESC
Sensors	GNSS, IMU, LiDAR (optional), Vision
Flight Controller	PX4 / ArduPilot + NeuroFC Ready
Compliance	DGCA Compliance-Ready BVLOS Capable (Testing)

KEY FEATURES

- ▶ Hybrid propulsion: vertical takeoff with fixed-wing forward efficiency
- ▶ Modular payload bay — camera, delivery pod, and sensors hot-swappable
- ▶ AI-assisted flight stabilization via NeuroFC integration
- ▶ Real-time telemetry and live mission monitoring
- ▶ Weatherproof airframe engineered for harsh field deployment

USE CASES

- ▶ Medical & Vaccine Delivery
- ▶ Defense Reconnaissance & Surveillance
- ▶ Precision Agriculture Monitoring
- ▶ Topographic Mapping & Surveying
- ▶ Disaster Relief Logistics
- ▶ Border Patrol & Area Security